

Insurance Fraud Detection Using Machine Learning

This project builds an end-to-end machine learning pipeline to detect fraudulent vehicle insurance claims for Global Insure, enabling early identification of high-risk claims before payouts are made. The analysis covers 1,000 historical claims described by 40 features spanning policy details, insured demographics, incident attributes, and claim amounts, with a naturally imbalanced target: only 24.7% of claims (247 of 1,000) were fraudulent.

Data preparation addressed real-world messiness: placeholder '?' values affected `property_damage` (360 records), `police_report_available` (343), and `collision_type` (178), while a fully empty column was dropped entirely. Exploratory analysis surfaced powerful fraud signals — fraudulent claims average \$60,302 against \$50,289 for legitimate ones, roughly 20% higher, and incident severity proved decisive: Major Damage incidents carry a 60.5% fraud rate compared with just 10.7% for Minor Damage and 6.7% for Trivial Damage. Engineered features such as `days_between_policy_and_incident` further exposed suspicious timing patterns.

To counter class imbalance, `RandomOverSampler` was applied to the training data, and `RFECV` with five-fold stratified cross-validation identified the most predictive features, ranking incident severity indicators at the top. Two models were developed. A Logistic Regression baseline, refined through sensitivity–specificity and precision–recall cutoff analysis, achieved 81.0% validation accuracy with 67.6% recall and 85.4% specificity. A Random Forest was then optimised via `GridSearchCV` across 576 hyperparameter combinations (2,880 total fits), selecting 100 estimators with a maximum depth of 20 and balanced subsample class weights, and delivering approximately 90.6% average cross-validation accuracy on resampled training data.

On unseen test data, the tuned Random Forest achieved 79.0% accuracy, 55.4% recall, 57.8% precision, 86.7% specificity, and an F1 score of 0.566 — a substantial improvement over the untuned model, whose recall rose from 0.27 to 0.55 and F1 from 0.30 to 0.56 through hyperparameter tuning. Threshold analysis showed that lowering the default 0.5 cutoff can further raise recall where the cost of missed fraud outweighs false alarms.

The business value is direct: automated fraud risk scoring lets investigators prioritise high-risk claims, accelerates approval of genuine ones, reduces financial leakage from high-value suspicious claims, and provides the interpretable, auditable decision framework insurers need for regulatory compliance.